Information Retrieval & Web Analytics Deliverable 1 Report

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GitHub URL: https://github.com/PaolaMartin01/IRWA_FinalProject
GitHub TAG:
https://github.com/PaolaMartin01/IRWA_FinalProject/releases/tag/IRWA-2026-part-1

Overview of the Dataset

The dataset consists of 28,080 fashion products and 17 attributes, which includes product titles, descriptions, brands, categories, product details, prices, discounts, average ratings, stock availability and more. It was then loaded into a Pandas Dataframe for further processing. Understanding the data is a crucial first step in preparing it for a search system.

We inspected the dataset for missing values and duplicates, reviewing data types, and verifying that expected columns were actually present. The results revealed that all product IDs were unique, and all expected columns were found. However, several fields contained missing or empty values, for example, 11,150 products had empty descriptions, 2,009 had empty brand values, and 1,643 had empty seller values. This can affect how a product is found, since a product without a description may be harder to retrieve through descriptive search terms. But it could still be found using its title or other available attributes. Therefore, missing values should be handled according to the purpose of each field rather than automatically removing entire products.

Data Preparation

The dataset was then text preprocessed, starting with flattening the product_details list, which converts the dictionaries into a single text string. This helps with further preprocessing. Afterwards, we decoded the HTML entities into normal characters, and removed HTML tags. Lowercase conversion was applied to make text more consistent followed by tokenization. Continuing with the preprocessing, stopwords were removed to reduce common words that can make the data difficult to process. However we kept important words such as “men”, “women”, “no”, and so on, as these can help distinguish products. Stemming was then applied to help identify words with a common stem. We then converted some columns into numerical values, for example the selling_price, actual_price and discount, to make them suitable for further calculations. However, not all values were successfully converted, in which case, were replaced with NaN. Finally we then organised the products using their unique pid (product_id), which helps further indexing.

Exploratory Data Analysis

Afterwards, we try to gain a better understanding of the dataset by analysing it. First, we count the total number of unique vocabularies, as well as the average title and description length. We found that the total number of unique vocabularies is 4896, the average title length is 6.27 tokens, while the average description length is 18.90 tokens. According to Figure 1, it is shown that the mode of title length is around 6 words which is similar to its mean.

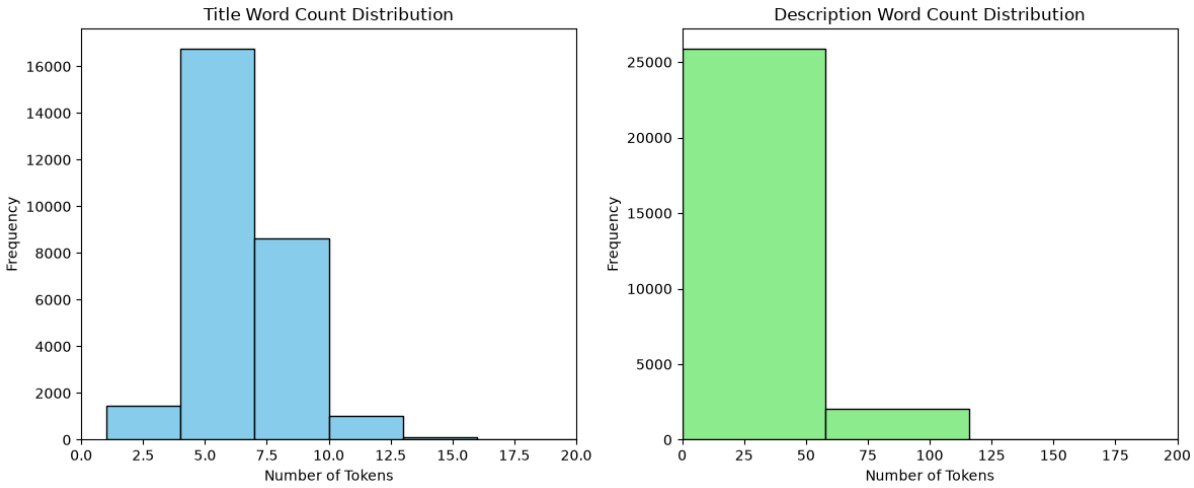


Figure 1: Length Distribution of Title and Description

Next, to be able to see what words are used in the title and description, we generate a word cloud as shown in Figure 2. These 2 word clouds reveal that the words ‘round’ and ‘shirt' are mentioned many times in the title, demonstrating that many items are round shirts. Moreover, the words ‘men’ and ‘women’ are also top words in the title, showing that title usually includes the intended sex of the item. Moving the focus to the description word cloud, ‘made’ and ‘cotton’ are popular. This indicates that most items are made of cotton. Polo remains a popular style of clothes as it is also a top word mentioned in the description.



Figure 2: Word Cloud of Title and Description

Figure 3 shows the distribution of selling price, discount percentage and average rating of products. The selling price histogram shows a strongly right-skewed pattern. The vast majority of products are concentrated in the lower price tiers (between 0 and 1,500), which suggests a predominantly budget-friendly or mass-market inventory. The discount percentage graph reveals that products are discounted at vastly different rates, but most products feature discounts between 30% and 70%, with the absolute highest frequency of discounts peaking around the 50% to 60% range. The average rating distribution is left-skewed, indicating that the majority of products are rated highly by consumers. Ratings are heavily concentrated between 3.5 and 4.5, with a distinct, massive spike precisely around the 3.5 mark. Very few products suffer from poor ratings below 2.5.

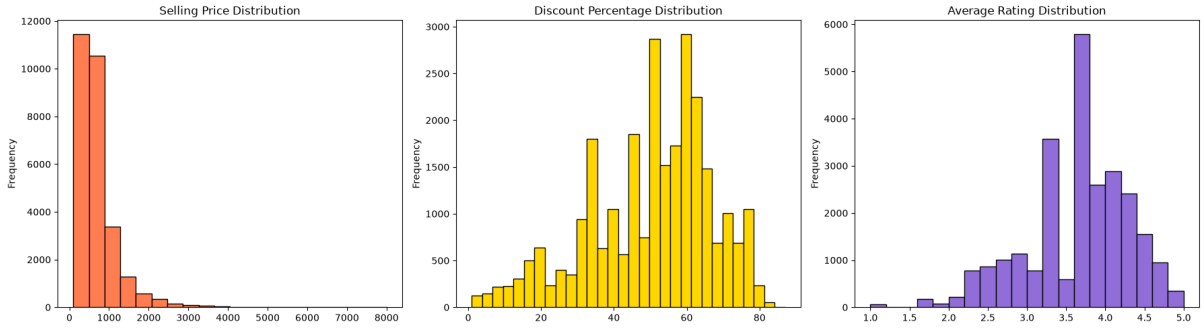


Figure 3: Distribution of Selling Price, Discounted Percentage, and Average Rating

Figure 4 presents that most of the products are available to buy with only less than 6% are out of stock.

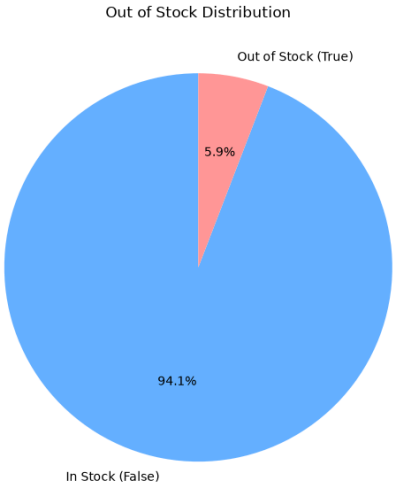


Figure 4: Out of Stock Distribution

Figure 5, 6, and 7 show brand and seller rankings. According to the "Top 10 Sellers by Product Count" chart, RetailNet dominates the platform by a wide margin, accounting for over 1,400 products. The rest trails significantly, with the second-highest seller, SandsMarketing being behind by over 500 products. The "Top 10 Brands by Product Count" chart demonstrates a much more gradual step-down compared to the seller distribution. ECKO Unl is the most represented brand with nearly 1,000 products. It is followed steadily by Free Authori, ARBO, and REEB. The "Top 10 Highest Rated Brands (Min 10 Products)" chart shows that VARTe leads the platform with a perfect 5.0 average rating. It is followed by other brands with average ratings falling between approximately 4.4 and 4.6.

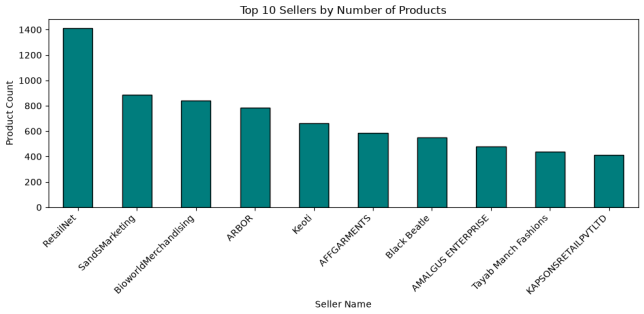


Figure 5: Top 10 Sellers by Product Count

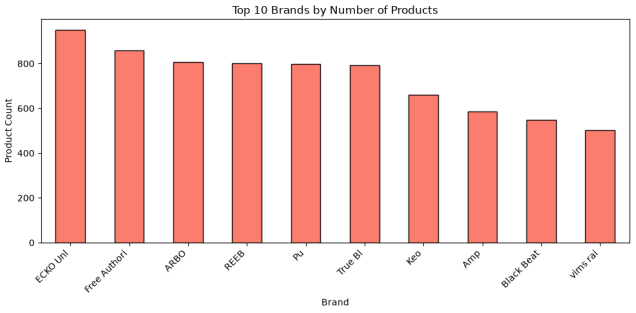


Figure 6: Top 10 Brands by Product Count

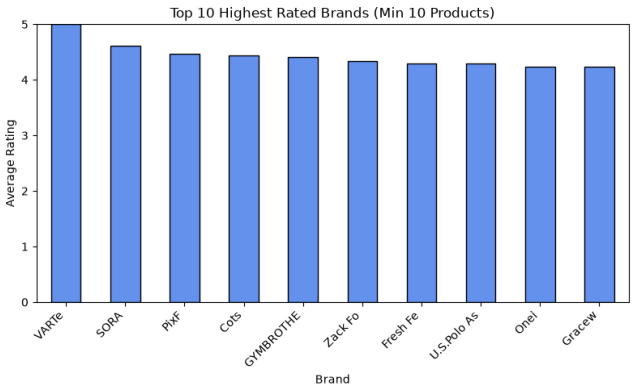


Figure 7: Top 10 Highest Rated Brands

Next, we also analyse the top highest rated products. According to Table 1, the highest rated products tend to be casual wear like T-shirts with cheaper prices as all are less than 1000. The most expensive items on the platform (Table 2) are heavily concentrated in outerwear (tracksuits, jackets, sweatshirts) and jeans, with prices tightly clustered just below the 8,000 mark. This might indicate that outerwear is harder to produce, making them more expensive. The heaviest discounts (Table 3) range from 84% to 87% off and are applied primarily to t-shirts. These suggest a strategy to clear out high-margin, seasonal, or excess inventory, dropping items that originally cost around 1,000 to drastically low prices.

Table 1: Top 5 Highest Rated Products

Title	**Average Rating**	**Selling Price**
Solid Women Round Neck Blue T-Shirt	5.0	385.0
Printed Women Hooded Neck Black T-Shirt	5.0	549.0
Printed Women Hooded Neck Grey T-Shirt	5.0	909.0
Graphic Print Men Round Neck Blue T-Shirt	5.0	519.0
Solid Women Round Neck White, Black T-Shirt	5.0	649.0

Table 2: Top 5 Most Expensive Products

PID	Title	Selling Price	Brand
TKTFZ3YGGMMNBYEZ	Solid Women Track Suit	7,999	REEB
BZRFVAX2QGTEGHRH	Checkered Single Breasted Party Women Full Sle...	7,998	True Bl
JCKFQF5KMJJ349H8	Full Sleeve Solid Women Casual Jacket	7,799	Pu
SWSFUMFGQFKVZGYH	Full Sleeve Printed Men Sweatshirt	7,799	Pu
JEAF8S4GWU5YKQTF	Skinny Men Blue Jeans	7,794	G

Table 3: Top 5 Heaviest Discounted Products

PID	Title	Discount	Actual Price	Selling Price
TSHF5FRXKGF6A4FH	Printed Women Round Neck White T-Shirt	87% off	998	128
TSHFMFXGFJ7G2ABK	Printed Women Round Neck Grey T-Shirt	86% off	999	136
TSHFMFT7VASAHBH3	Printed Women Round Neck White T-Shirt	86% off	999	136
TSHFGH6T3CVGDXS9	Printed Men Round Neck Multicolor T-Shirt (Pa...	85% off	2,999	449
CAPE9YWMURE4FKAJ	Solid Balclava Cap	84% off	1,499	228

AI Usage

We use AI to review the code written and help correct some grammar in the report, as well as the README file.